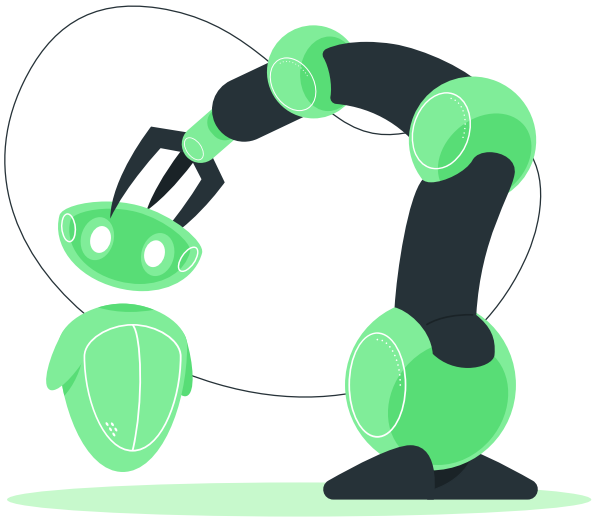
Robotics CS 3015

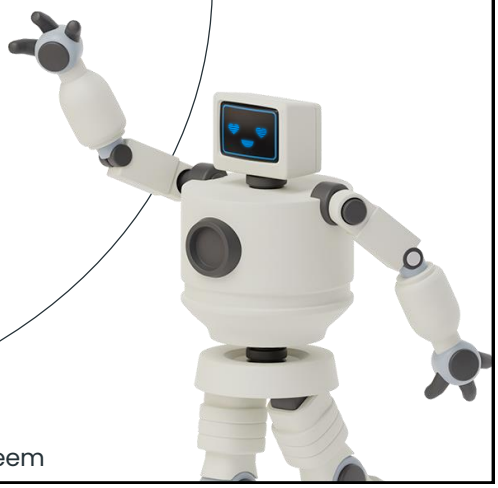
Umarah Qaseem



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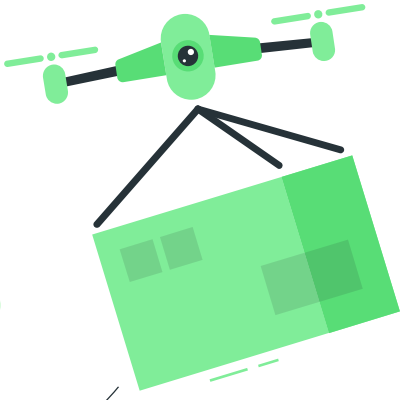
1 b

- Introduction to Robotics
- Alternative Taxonomy
- Working of Some famous robots
- Basics of a Robot
- Key terms
- Hardware of a Robot
- Software for Robotics



2

Home Tasks?

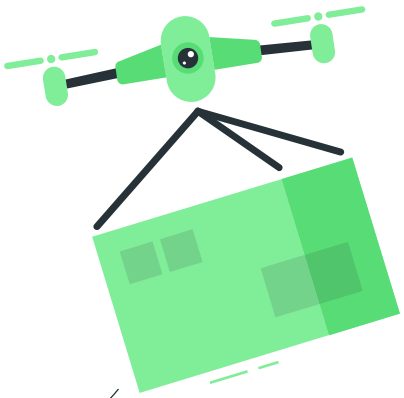


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What is Robotics?

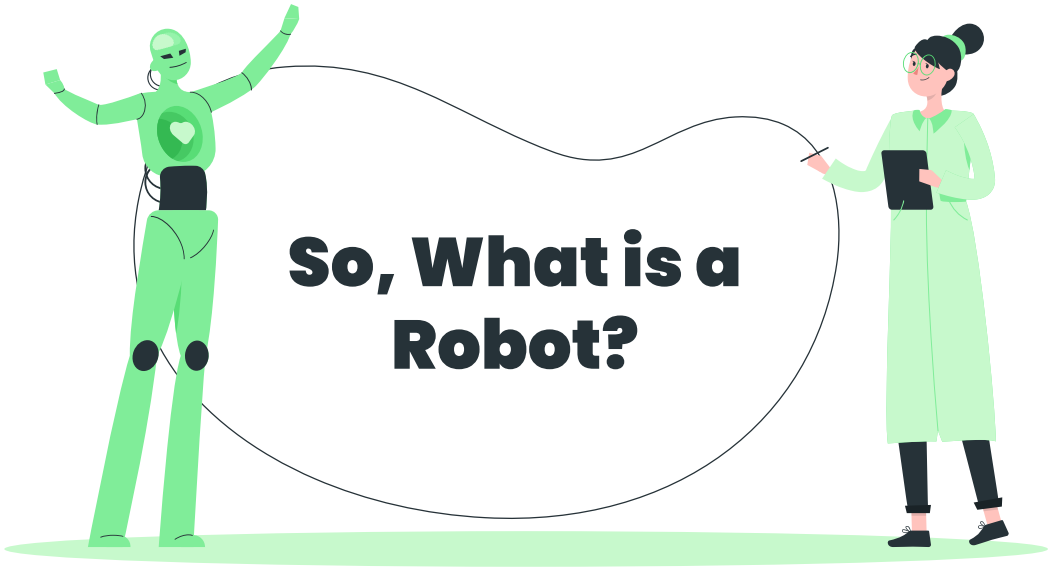
Robotics is an interdisciplinary field that deals with the study of Robots' control, sensory feedback and information processing.

Robots are used to perform tasks done traditionally by human beings.



Revision

4



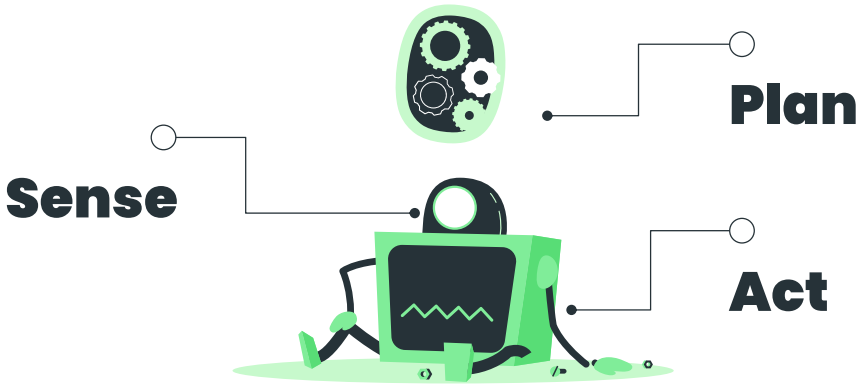
5

What is a Robot?

- “a goal-oriented machine that can **sense, plan and act**”
- A robot senses its environment and uses that information, together with a goal, to plan some action.

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Key Terms



7

Key Terms



A machine that moves things from A to B



A machine that moves from A to B

A goal oriented machine that can sense, plan and act

8

Sensing

- Exteroceptive sensors and Proprioceptive sensors

- **Proprioceptive sensors:** measure the state of the robot itself.

Examples:
- the angle of the joints on a robot arm,
- the number of wheel revolutions on a mobile robot
- the current drawn by an electric motor.
- battery voltage

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Sensing

- **Exteroceptive sensors:** measure the state of the world with respect to the robot.

Examples:
- The sensor might be a simple bump sensor on a vacuum cleaner robot to detect collision.
- It might be a GPS receiver that measures distances to an orbiting satellite constellation,
- A compass that measures the direction of the Earth's magnetic field vector relative to the robot.
- It might also be an active sensor that emits acoustic, optical or radio pulses in order to measure the distance to points in the world based on the time taken for a reflection to return to the sensor.

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What about cameras?

Exteroceptive or Proprioceptive?

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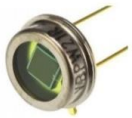
Sensors

- Sensors are nothing but measuring instruments which measures quantities such as position, velocity, force, torque, proximity, temperature, etc.


Temperature Sensor


Humidity Sensor


Proximity Sensor


Light Sensor


Metal Sensor


Color Sensor

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Sensors

- Contact sensors
  - Push buttons and contact switches
  - Pressure sensors
- Distance/Proximity sensors
  - Ultrasonic range finders,
  - Infrared sensors,
  - Laser measurement devices and sensors
- Positioning sensors
  - RFID,
  - Room navigation (indoor localization) sensors
  - GPS and other live tracking devices

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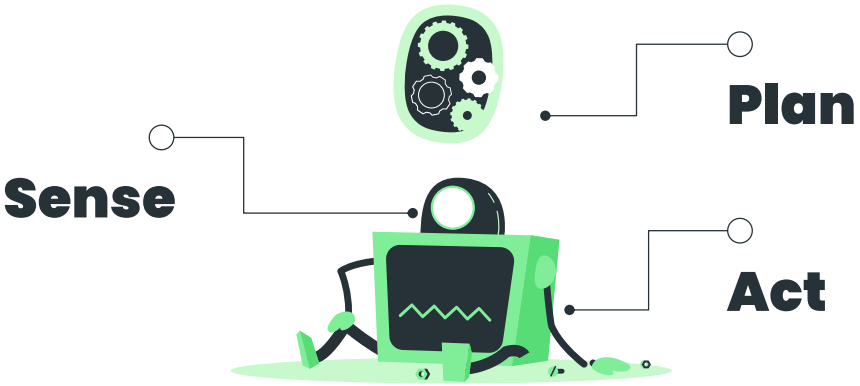
Sensors

- Voltage sensor
- Rotation sensors
  - Motion sensors/Potentiometers
  - Gyro sensors (rotational speed)
- Environmental sensors
  - Light sensors, Sound sensors
  - Temperature, flame, smoke, Alcohol sensors,
  - Thermal imaging cameras
  - Humidity and moisture sensors
  - Pressure sensors, Gas sensors
- Biometric and Medical Sensors

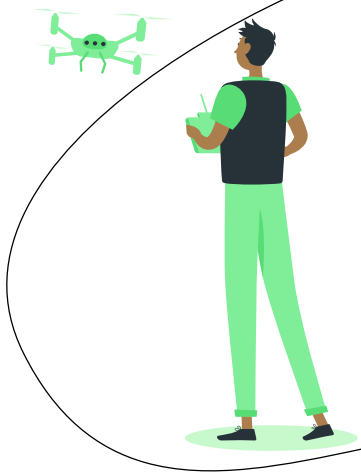
Fingerprint, Heart Rate, Blood Oxygen level, Blood pressure

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Key Terms

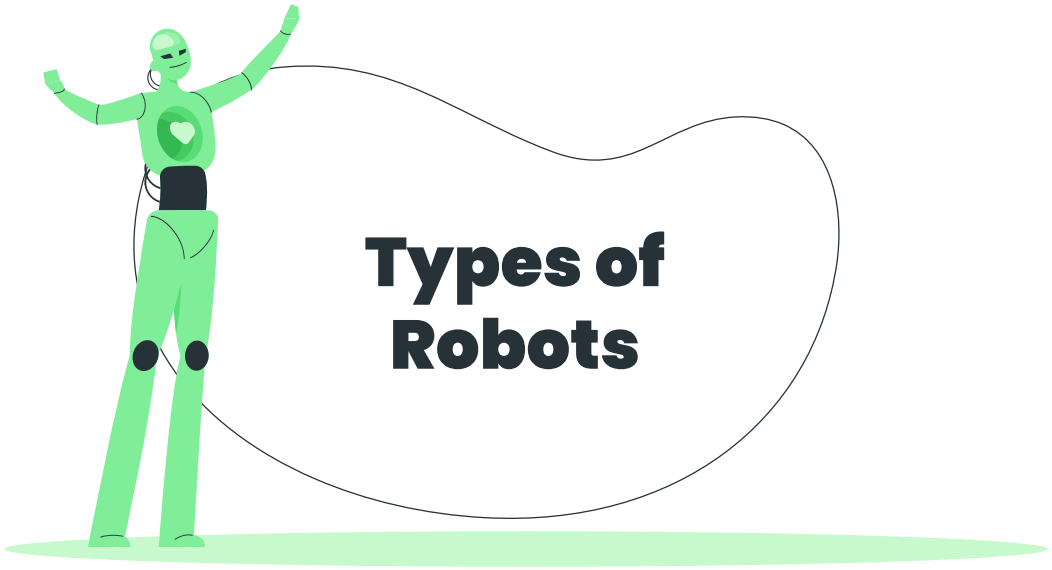


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- Plan?
- Act?

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Types of Robots

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Types of Robots

- Industrial/Stationary
- Wheeled (Mobile Robots)
- Legged (Mobile Robots?)
- Autonomous Underwater Vehicle (AUV)
- Unmanned Aerial Vehicle (UAV)
- Swarm
- Nano
- Soft robots
- Androids
- Bots/agents/RPA?

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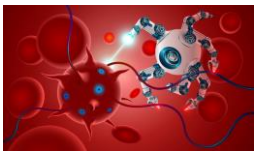










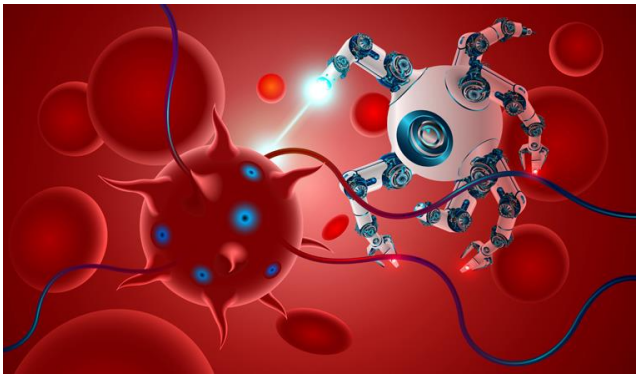


Robotics – Umarah Qaseem

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Nano Robots

- Nano robots are used for drug delivery.
- Normally drugs work through entire body before they reach effective area.



Robotics – Umarah Qaseem

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Androids

- Robots can but not necessarily be in human forms, Androids are always in human forms.



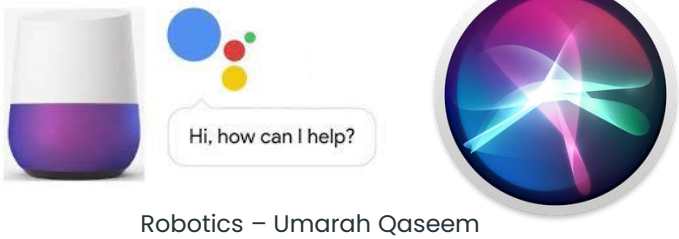

Soft Robots

- Soft robots are made from soft materials.
- Materials similar to those found in living organisms
- They adapt heavily from the way in which living organisms move and adapt to their surroundings.
- Applications: Biomedical, surgery, space exploration and prosthetics (replace severed limbs)
- Bots – software – Robotics Processes Automation. Software that do automatized repetitive tasks.

Bots

- Bots are software robots
- Also called as agents
- **RPA** (Robotics Processes Automation):
  Software that automate repetitive tasks.

- Examples?
- Virtual Assistants (Siri), Chatbots, expert systems, web scrapers, web crawlers



RPA

- Robotics Processes Automation is a technology based on metaphorical software robots (bots). It is sometimes referred to as software robotics (not to be confused with robot software).



SWARM Robotics

- Robots as a group
- Applications: Mining, agricultural foraging, medicinal nano robots, military etc.



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Alternative Taxonomy

Based on Robots' Function



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Alternative Taxonomy

- **Manufacturing Robots**
- Function:
  operate in factories and are the technological descendants of the first-generation robots.
- Application:
  factory automation, packaging, electronic assembly, welding, painting etc.

Alternative Taxonomy

- **Service Robots:**
- Function:
  supply services to people
- Applications:
  medical rehabilitation, fetching, carrying, cleaning, personal assistance etc.

Alternative Taxonomy

- **Field Robots**
- Function:
  work on outdoor tasks
- Applications:
  environmental monitoring, agriculture, mining, construction, forestry, agricultural robots, driverless cars etc

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Alternative Taxonomy

- **Humanoid Robots**
- have the physical form of a human being – they are both mobile robots and service robots

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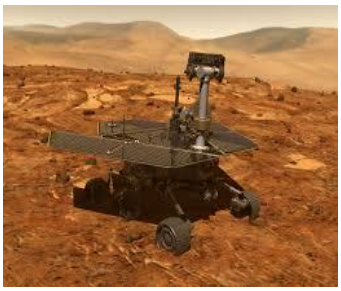
Alternative Taxonomy

- **Telerobots**
- are robot-like machines that are remotely controlled by a human operator. Davinci surgical robot, mars rover etc



Davinci Surgical System

(Courtesy: Intuitive Surgical)



Mars Rover

(Courtesy: NASA)



Hardware of Robots

Hardware

- Frame
- Controller
- Manipulators
- Actuators
- End Effectors
- Sensors
- Motors
- Power Supply

Body or Frame

- The body of robot can be of any shape and size. Essentially, the frame (or body) provides the structure of the robot.
- Majority of Robots look nothing like humans (as shown in movies).
- Robots are designed more based on their function than their appearance.

Controller

- The digital computer (both the hardware and the software) acts as a controller to the robot. The controller functions in a manner analogous to the human brain.
- With the help of this controller, the robot can carry out the assigned tasks.
- The controller directs and controls the movement of the Robotic arm and the End effector.
- Examples?

Motors

- The basic types of robotic motors available include:
- AC or DC motors
- They are useful for very high torque (AC motors) or high speed (DC motors) applications.
- Which kind of motors are usually used with mobile robots?

Motors – Servo motors

- An AC, DC or brushless DC motor combined with a position-sensing device and a control circuit.
- It is a rotary actuator that allows for precise control of angular position. The motor is attached by gears to the control wheel, and as the motor rotates, the position sensors resistance changes, so the control circuit can precisely regulate movement.
- Often combined with a servo drive control, a type of electronic amplifier used to power and monitor the feedback from servomechanisms.

Motors – Stepper motors

- A brushless DC electric motor that divides a full rotation into a number of equal steps.
- The motor's position can then be made to move and hold at one of these steps without any position sensor for feedback.
- With a computer-controlled stepping, these motors can be used to achieve very precise positioning and/or speed control, ideal in very numerous robotics applications.

Manipulator – Robotic Arm

- Robotic Manipulator is also called as robotic arm.
- A series of joint segments are put together to form an arm-like manipulator that is capable of automatically moving objects.
- Unlike the frame and the Control System, manipulators are not integral to a robot, i.e., a robot can exist without a manipulator
- The industrial robots usually just have manipulator and a Fixed base.

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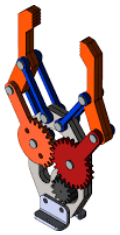


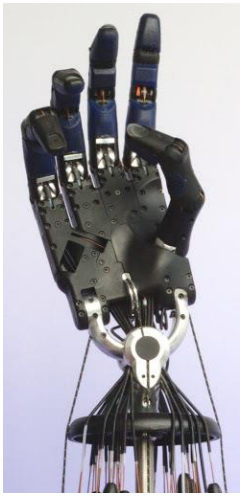
45

End effectors

- End effector is the device at the end of a robotic arm, designed to interact with the environment.
- The exact nature of this device depends on the application of the robot.
- Robot Hand







Actuators

- Actuation is conversion of energy to mechanical energy.
- Actuators are used to produce mechanical energy in robots.
- Actuators are muscles of robots.
- E.g., motors
- It translates control signals into mechanical movement.



Actuators – The Locomotion Device

- In Human Beings, the power for the movement of the arm, the palm and fingers is provided by muscles.
- In robots the power for the locomotion is provided by the **motors**.
- The motors used for providing locomotion in robots are of three types depending on the source of energy
  - Electric
  - Hydraulic
  - Pneumatic

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Power Supply

What about it?

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Arduino

- An **Arduino** board is composed of a microcontroller, some LEDs, a reset button, and many pins that you can **use** for input/output operations.
- Pins help to easily read data from sensors, or control different motors and actuators.



Arduino Micro



Arduino Uno R3



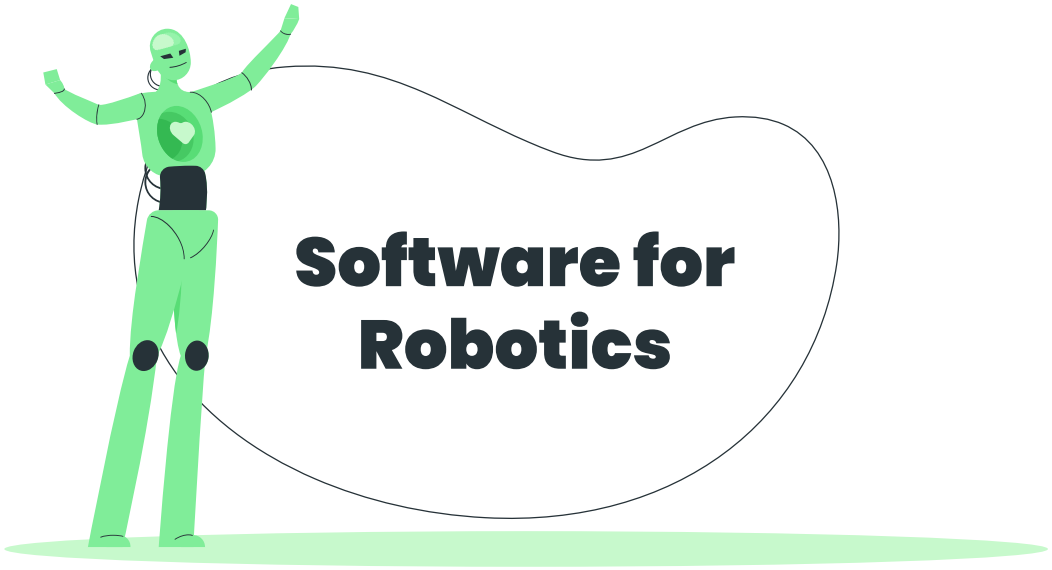
Arduino MEGA 2560 R3 (Made in Italy)

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Raspberry pi



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Software for Robotics

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ROS

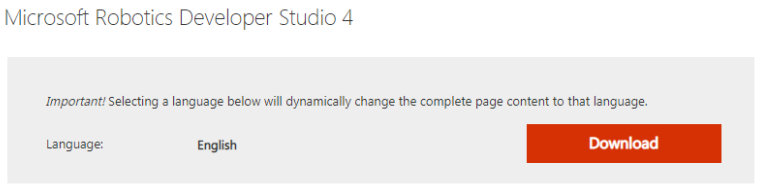
- **Robot Operating System**
- Provides libraries and tools to help software developers create robot applications.
- Provides hardware abstraction, device drivers, libraries, visualizers, message-passing, package management, and more.
- ROS is licensed under an open source, BSD license.
- http://wiki.ros.org



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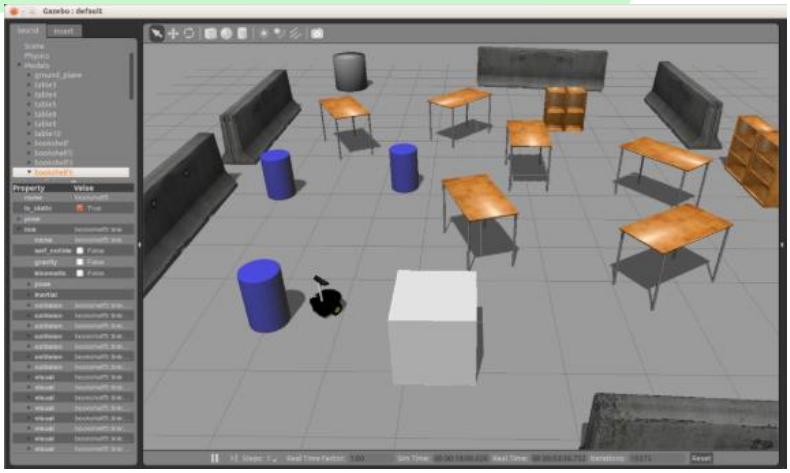
Integrated Development Environments

- Webot
- Rviz (C++, Python)
- Vrep (C++)
- Gazebo
- Moveit
- MRDS (Microsoft Robotic Development Studio)
- And many more.



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Gazebo



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Home Task

Watch the following movie before coming in next class

Spare Parts 2015



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